

Aditya Kumar Pattanayak

Software Development — Backend Systems — Machine Learning & AI

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Education

- **International Institute of Information Technology, Bhubaneswar** *Aug 2023 – July 2027 (Expected)*
Bachelor of Technology in Information Technology **CGPA: 7.87/10.0**
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Probability & Statistics, Linear Algebra
- **DAV Unit 8, Bhubaneswar** *Jun 2020 – May 2022*
Senior Secondary (XII) Percentage: **94.8%**

Experience

- **Backend Engineer** *Oct 2025 – Jan 2026*
Stealth Startup (Remote)
 - Built 0→1 AI-powered market and founder-intelligence platform serving 100+ clients: real-time signal feeds, LinkedIn RAG vector indexing, automated company research reports.
 - Architected AWS infra (ECS, ECR, S3), Apify enrichment pipelines, and full-stack React/TypeScript + backend; Dockerized services with CI/CD for production reliability.
 - Integrated third-party APIs, Stripe subscriptions, and data aggregation workflows handling production-scale traffic.

Projects

- **Tokaroo - Adaptive RAG Evaluation & Optimization Framework** 🌐
Python, FastAPI, React, TypeScript, SentenceTransformers, Cross-Encoders
 - Designed and built evaluation systems measuring Recall@K, MRR, nDCG, Hit Rate, Context Utilization, and Answer Quality across 65+ retrieval configurations.
 - Engineered hybrid retrieval pipelines (Hybrid Search + HyDE + RRF Fusion + Cross-Encoder Reranking) enabling systematic comparison of strategies and diagnosis of 10+ real-world failure modes.
 - Delivered a production-ready evaluation platform used to optimize retrieval performance and system reliability.
- **Aegis - AI Agent Reliability Runtime** 🌐 [Link]
Python, FastAPI, OpenTelemetry, Grafana, Docker
 - Designed and deployed a middleware-style reliability layer between LLM planners and external tools, enforcing deterministic capability, argument, confidence, and response validation with graceful recovery.
 - Built an evaluation harness measuring correctness, consistency (Pass@K/Pass^k), failure modes, and p50/p95 latency against an unguarded baseline; reduced silent failures from 6 → 0 in an initial 22-case benchmark.
 - Added OpenTelemetry + Grafana distributed tracing across planning, validation, execution, and recovery, enabling stage-level latency diagnosis and operational debugging.

Technical Skills

Languages: Python, C++, Go, TypeScript, JavaScript, SQL

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Complexity Analysis, Problem Solving

Backend & Systems: FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, Redis

Cloud & Infrastructure: AWS (ECS, ECR, S3), Docker, GitHub Actions, CI/CD

ML Frameworks: PyTorch, TensorFlow.js, Scikit-Learn, SentenceTransformers

Frontend: React, Next.js, TailwindCSS

Achievements

- **Winner** – ETHMumbai Triple Track Challenge for PayGate.
- **Winner** – Dev Challenge Hackathon; Hivecraft featured as Project of the Week on Peerlist (200+ users).
- **2nd Runner-Up** – COMPOSIT Case Study Competition, IIT Kharagpur.
- **5th Place** – Datamatics Hackathon, IIT Bombay (among 156 teams).
- **Top 3 Category Winner** – AmpliCode Hackathon (13th overall among 925 participants).